OBJECT ORIENTED PROGRAMMING

ACTIVITY #4 – Student Enrollment Assessment System

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Instruction:

1. Create a Java program named **EnrollmentAssessmentSystemLastName**.
2. Import and use the Scanner class to accept user input.
3. Ask the user to input the following information:
   - Student Name
   - Senior High School Strand (STEM, ABM, HUMSS, TVL, Others)
   - Entrance Exam Score (0–100)
   - Interview Score (0–100)
   - First Screening Result (Pass/Fail)
4. Compute the Final Evaluation Score using the formula:

$$FinalScore = (EntranceExam \times 0.60) + (InterviewScore \times 0.40)$$

5. Use a nested if statement to determine the admission result.
   Conditions:
   - If First Screening = Pass
     - If FinalScore ≥ 85 → Admitted with Scholarship
     - If FinalScore ≥ 75 → Admitted
     - If FinalScore ≥ 65 → Waitlisted
     - Otherwise → Not Qualified
   - If First Screening = Fail
     - Automatically Not Qualified
6. Use a switch statement to determine Program Recommendation based on the strand:

Strand	Recommended Program
STEM	BSIT / BSCS
ABM	BSBA
HUMMS	BSED / AB Communication
TVL	BSIT
Others	General Program

7. Display the result in the following format:

```
===== ENROLLMENT ASSESSMENT RESULT =====

Student Name: Maria Santos
SHS Strand: STEM
Final Evaluation Score: 82.50
Admission Result: Admitted
Recommended Program: BSIT / BSCS
```

Requirements:

- Scanner class
- Nested if statement
- Switch statement
- Proper variable declarations
- Computation using formulas (no hardcoding)

Source Code:

```java
import java.util.Scanner;

public class EnrollmentAssessmentSystemRivas {

    public static double calculateFinalScore(int entranceScore, int interviewScore) {
        // Formula to calculate
        double finalScore = (entranceScore * 0.60) + (interviewScore * 0.40);
        return finalScore;
    }

    public static String admissionResult(String firstScreening, double finalScore) {
        // Converts to lowercase to ensure format
        firstScreening = firstScreening.toLowerCase();

        // Returns String message depending on the conditions
        if (firstScreening.equals(anObject: "pass")) {
            if (finalScore >= 85) {
                return "Admitted with Scholarship";
            } else if (finalScore >= 75) {
                return "Admitted";
            } else if (finalScore >= 65) {
                return "Waitlisted";
            } else {
                return "Not Qualified";
            }
        } else if (firstScreening.equals(anObject: "fail")) {
            return "Not Qualified";
        } else {
            return "Invalid Input!";
        }
    }

    public static String programRecommendation(String shsStrand) {
        // Converts to uppercase to ensure format
        shsStrand = shsStrand.toUpperCase();

        // Returns String message
```

```java
        // Variable initialization
        String studentName;
        String shsStrand;
        int entranceScore;
        int interviewScore;
        String firstScreening;

        System.out.println(x: "===== ENROLLMENT ASSESSMENT SYSTEM =====\n");
        System.out.print(s: "Enter your Name: ");
        studentName = input.nextLine();
        System.out.print(s: "Enter your Senior High School Strand: ");
        shsStrand = input.nextLine();
        System.out.print(s: "Enter your Entrance Exam Score: ");
        entranceScore = input.nextInt();
        System.out.print(s: "Enter your Interview Score: ");
        interviewScore = input.nextInt();
        input.nextLine(); // Consume the newline left in the buffer
        System.out.print(s: "Enter your result for the first screening (Pass/Fail): ");
        firstScreening = input.nextLine();

        // Calls every function and assign to varaible for other function to use
        double finalScoreResult = calculateFinalScore(entranceScore, interviewScore);
        String admissionMessage = admissionResult(firstScreening, finalScoreResult);
        String programResult = programRecommendation(shsStrand);
        // Calls display result function
        displayResult(studentName, shsStrand, finalScoreResult, admissionMessage, programResult);

        input.close();
    }
}
```

```java
        // Returns String message
        switch (shsStrand) {
            case "STEM":
                return "BSIT / BSCS";
            case "ABM":
                return "BSBA";
            case "HUMSS":
                return "BSED / AB Communication";
            case "TVL":
                return "BSIT";
            case "OTHERS":
                return "General Program";
            default:
                return "Invalid Strand!";
        }
    }

    public static void displayResult(String Name, String Strand, double FinalEval, String AdmissionResult,
            String RecoProgram) {

        // Displays everything
        System.out.println(x: "\n===== ENROLLMENT ASSESSMENT RESULT =====\n");
        System.out.println("Student Name: " + Name);
        System.out.println("SHS Strand: " + Strand);
        System.out.println("Final Evaluation Score: " + FinalEval);
        System.out.println("Admission Result: " + AdmissionResult);
        System.out.println("Recommended Program: " + RecoProgram);
    }

    Run | Debug
    public static void main(String[] args) {
        // Declare input scanner
        Scanner input = new Scanner(System.in);

        // Variable initialization
```

User Input:

```
===== ENROLLMENT ASSESSMENT SYSTEM =====

Enter your Name: Dave Aillerr Rivas
Enter your Senior High School Strand: STEM
Enter your Entrance Exam Score: 98
Enter your Interview Score: 75
Enter your result for the first screening (Pass/Fail): Pass
```

Final Output:

```
===== ENROLLMENT ASSESSMENT RESULT =====

Student Name: Dave Aillerr Rivas
SHS Strand: STEM
Final Evaluation Score: 88.8
Admission Result: Admitted with Scholarship
Recommended Program: BSIT / BSCS
```

Program Process:

```
===== ENROLLMENT ASSESSMENT SYSTEM =====

Enter your Name: Dave Aillerr Rivas
Enter your Senior High School Strand: STEM
Enter your Entrance Exam Score: 98
Enter your Interview Score: 75
Enter your result for the first screening (Pass/Fail): Pass

===== ENROLLMENT ASSESSMENT RESULT =====

Student Name: Dave Aillerr Rivas
SHS Strand: STEM
Final Evaluation Score: 88.8
Admission Result: Admitted with Scholarship
Recommended Program: BSIT / BSCS
```